Midterm Review Questions
Weeks 1 - 7

Week 1: Kernel, Processes, and Threads

1. **What is the primary function of a system call in an operating system?** *(Slide: System Calls)*
    - A) To allow users to install new applications
    - B) To enable direct hardware access for user programs
    - C) To provide an interface between user programs and the operating system
    - D) To increase CPU clock speed
    - Answer: C
2. **Which system call is used to create a child process in UNIX-based operating systems?** *(Slide: fork() System Call)*
    - A) exec()
    - B) clone()
    - C) fork()
    - D) create_process()
    - Answer: C
3. **What is the key advantage of using threads over processes?** *(Slide: Thread Model)*
    - A) Threads can run on different operating systems simultaneously
    - B) Threads require less overhead and can be created/destroyed faster than processes
    - C) Threads do not require synchronization
    - D) Threads have separate memory spaces
    - Answer: B
4. **Which component of an operating system is responsible for managing processes?** *(Slide: Process Model)*
    - A) File system
    - B) Kernel
    - C) Memory manager
    - D) Scheduler
    - Answer: D
5. **What does the `execve()` system call do?** *(Slide: System Calls for Process Management)*
    - A) Creates a new process
    - B) Terminates a process
    - C) Replaces the current process image with a new program
    - D) Waits for a child process to terminate

- Answer: C

Week 2: Interprocess Communication and Synchronization

6. **What is a race condition in the context of interprocess communication?** *(Slide: Race Conditions)*
    - A) A condition where processes run in parallel without synchronization
    - B) A situation where multiple processes access shared data simultaneously, leading to unpredictable results
    - C) A scheduling algorithm that prioritizes faster processes
    - D) A security flaw allowing unauthorized access to memory
    - Answer: B
7. **What synchronization mechanism prevents multiple processes from entering a critical section simultaneously?** *(Slide: Semaphores)*
    - A) Virtual Memory
    - B) Mutex
    - C) Semaphore
    - D) Deadlock
    - Answer: C
8. **Which of the following is true about mutual exclusion?** *(Slide: Critical Region)*
    - A) Multiple processes can access shared data at the same time
    - B) Only one process can be in the critical section at a time
    - C) Processes never need synchronization
    - D) Critical regions can be accessed by all threads without restriction
    - Answer: B
9. **Which process synchronization method allows one process to wait for a specific condition before continuing?** *(Slide: Conditional Variables)*
    - A) Deadlock prevention
    - B) Condition variables
    - C) Spinlock
    - D) Shared memory
    - Answer: B
10. **What is the purpose of Peterson's Algorithm?** *(Slide: Peterson's Algorithm)*
    - A) Process scheduling
    - B) Implementing page replacement
    - C) Achieving mutual exclusion in concurrent processes
    - D) Memory allocation

- Answer: C

Week 3: Scheduling and Process Management

11. **Which scheduling algorithm assigns each process a fixed time quantum?** *(Slide: Round Robin Scheduler)*
    - A) First Come, First Serve
    - B) Shortest Job First
    - C) Round Robin
    - D) Priority Scheduling
    - Answer: C
12. **Which scheduler is responsible for determining which process to run next?** *(Slide: Scheduling Processes)*
    - A) Long-term scheduler
    - B) Short-term scheduler
    - C) Medium-term scheduler
    - D) Process manager
    - Answer: B
13. **What does the Xinu 'readylist' store?** *(Slide: Process Table - Readylist)*
    - A) Currently running processes
    - B) Processes waiting for I/O
    - C) Processes ready to execute but not currently running
    - D) Completed processes awaiting termination
    - Answer: C
14. **What is the main disadvantage of First Come, First Serve (FCFS) scheduling?** *(Slide: First Come First Serve)*
    - A) It requires complex hardware support
    - B) Shorter processes can get stuck waiting for longer processes
    - C) It does not work for batch systems
    - D) It is not preemptive
    - Answer: B
15. **Which scheduling algorithm gives priority to the process with the shortest execution time?** *(Slide: Shortest Job First)*
    - A) Round Robin
    - B) First Come, First Serve
    - C) Shortest Job First
    - D) Multilevel Queue
    - Answer: C

Week 4: Memory Management and Virtual Memory

16. **What is the purpose of the base and limit registers in memory management?** ***(Slide: Address Space - Base and Limit Registers)***

- A) To store the entire process's data
- B) To determine the maximum number of processes that can run
- C) To define the memory boundaries for a process
- D) To keep track of free memory pages
- Answer: C

17. **Which memory allocation strategy is the fastest but may cause fragmentation? *(Slide: Memory Allocation Algorithms)***
    - A) Best Fit
    - B) Worst Fit
    - C) First Fit
    - D) Next Fit
    - Answer: C
18. **What happens when a process tries to access a memory page that is not currently in RAM? *(Slide: Page Fault Handling)***
    - A) The OS loads the required page from disk into memory
    - B) The process is terminated immediately
    - C) The memory is compacted to make room for the page
    - D) The CPU restarts the entire system
    - Answer: A
19. **What is the main advantage of using paging over contiguous memory allocation? *(Slide: Paging)***
    - A) Paging eliminates internal fragmentation completely
    - B) Paging avoids external fragmentation by allocating memory in fixed-size blocks
    - C) Paging allows processes to share memory dynamically
    - D) Paging is faster than contiguous memory allocation
    - Answer: B
20. **Which hardware component speeds up the translation of virtual addresses to physical addresses? *(Slide: Translation Lookaside Buffer - TLB)***
    - A) Memory buffer register
    - B) Cache memory
    - C) Translation Lookaside Buffer (TLB)
    - D) Page table
    - Answer: C

Week 5: Paging and Virtual Memory Management:

21. **What is the main goal of the Least Recently Used (LRU) page replacement algorithm? *(Slide: Least Recently Used Algorithm)***
    - A) To replace the oldest page in memory
    - B) To replace the page that has not been used for the longest time
    - C) To randomly replace a page from memory
    - D) To keep frequently used pages in disk storage
    - Answer: B

22. **What is a "page fault"? *(Slide: Page Fault)***
    - A) A type of memory corruption
    - B) When a process accesses a page that is not in physical memory
    - C) When the CPU fails to allocate enough memory
    - D) When two processes share the same page
    - Answer: B
23. **Which technique reduces the number of page faults by keeping frequently used pages in memory? *(Slide: Working Set Model)***
    - A) Demand Paging
    - B) Prepaging
    - C) Thrashing
    - D) Working Set Model
    - Answer: D
24. **What happens if a system has too many page faults? *(Slide: Page Fault Frequency - PFF Algorithm)***
    - A) It enters a thrashing state
    - B) It improves performance
    - C) The OS increases the time quantum
    - D) The system automatically clears cache memory
    - Answer: A
25. **What is the purpose of the "dirty bit" in a page table entry? *(Slide: Page Table - R and M Bits)***
    - A) It marks pages that need to be reloaded
    - B) It tracks whether a page has been modified and needs to be written to disk
    - C) It prevents pages from being swapped out
    - D) It indicates a page is no longer in use
    - Answer: B

Week 6: File Systems and Storage Management

26. **Which data structure is used to track file metadata, including ownership and disk location? *(Slide: I-Nodes)***
    - A) Page Table
    - B) File Allocation Table (FAT)
    - C) Inode
    - D) Linked List
    - Answer: C
27. **Which file allocation strategy links blocks together, allowing files to grow dynamically? *(Slide: Linked-List Allocation)***
    - A) Contiguous Allocation
    - B) Indexed Allocation
    - C) Linked-List Allocation
    - D) Direct Allocation
    - Answer: C

28. **What is the primary purpose of a file system?** ***(Slide: OS File Abstraction)***
    - A) To manage hardware
    - B) To structure disk storage and provide access to data
    - C) To optimize CPU scheduling
    - D) To allocate memory
    - Answer: B
29. **Which file organization method minimizes fragmentation and provides efficient random access?** ***(Slide: File System Implementations - Indexed Allocation)***
    - A) Contiguous allocation
    - B) Linked-list allocation
    - C) Indexed allocation
    - D) FAT allocation
    - Answer: C
30. **What is a journaling file system?** ***(Slide: Journaling File System)***
    - A) A file system that reduces disk fragmentation
    - B) A system that logs changes before applying them to prevent corruption
    - C) A real-time file compression system
    - D) A system that automatically encrypts file data
    - Answer: B

Week 7: I/O Systems and File-System Optimization

31. **What is the main advantage of memory-mapped I/O?** ***(Slide: Memory-Mapped I/O)***
    - A) It eliminates the need for RAM
    - B) It reduces CPU overhead by allowing direct access to I/O devices
    - C) It speeds up disk fragmentation
    - D) It prevents page faults
    - Answer: B
32. **What is the role of Direct Memory Access (DMA) in I/O operations?** ***(Slide: Direct Memory Access - DMA)***
    - A) It allows devices to communicate without CPU intervention
    - B) It prevents page faults
    - C) It synchronizes file writes
    - D) It increases cache size
    - Answer: A
33. **Which file system optimization technique improves read performance by caching disk blocks in memory?** ***(Slide: Buffer Cache)***
    - A) Defragmentation
    - B) Buffer Cache
    - C) Journaling
    - D) Quick Fit
    - Answer: B

34. **How does the operating system keep track of free disk space efficiently? *(Slide: Keeping Track of Free Blocks - Bitmap vs. Linked List)***
    - A) Using a bitmap or a free block linked list
    - B) By scanning the disk for empty spaces
    - C) By writing metadata to each block
    - D) By defragmenting the disk periodically
    - Answer: A
35. **Which scheduling algorithm is commonly used for disk I/O operations to minimize seek time? *(Slide: Disk Scheduling Algorithms - SCAN Algorithm)***
    - A) Round Robin
    - B) First Come, First Serve
    - C) SCAN (Elevator Algorithm)
    - D) FIFO
    - Answer: C

Free Response Questions

1. System Calls & Process Management
   Explain the role of system calls in an operating system. Describe how the `fork()` system call works and what happens when it is executed in a UNIX-based OS. Include an example scenario where `fork()` is useful.

2. Interprocess Communication & Synchronization
   Describe the concept of a race condition and explain how it can occur in a shared-memory system. What mechanisms, such as semaphores or mutexes, can be used to prevent race conditions? Provide an example.

3. Scheduling Algorithms
   Compare and contrast Round Robin (RR) and Shortest Job First (SJF) scheduling algorithms. Which situations are best suited for each, and what are the advantages and disadvantages of both?

4. Memory Management
   What is paging, and how does it differ from segmentation? Discuss the benefits of paging and explain how the Translation Lookaside Buffer (TLB) helps improve performance in a paging system.

5. Page Replacement Policies
   A computer system is running Least Recently Used (LRU) and First In, First Out (FIFO) page replacement algorithms. Explain how both algorithms decide which page to replace and analyze their effectiveness in different workloads.

6. File Systems & Storage
   Explain how inodes are used in a file system to track file metadata. What information is

stored in an inode, and how does it help in managing files efficiently?

7. **Virtual Memory & Thrashing**
   Define thrashing in the context of virtual memory. What causes thrashing, and what strategies can the operating system use to prevent it?

8. **I/O Systems & Direct Memory Access (DMA)**
   What is Direct Memory Access (DMA), and how does it improve system performance? Describe the steps involved in a DMA transfer and compare it with Programmed I/O (PIO).

9. **Disk Scheduling Algorithms**
   Describe how the SCAN (Elevator) scheduling algorithm works for disk I/O operations. Compare it to FCFS (First-Come, First-Serve) and LOOK scheduling in terms of performance and seek time efficiency.

10. **File System Consistency & Journaling**
    What is a journaling file system, and how does it help maintain file system consistency after a crash? Explain the process of journaling and why it is preferred over traditional file system recovery techniques.